

Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	4 (Term 1)	**Year:**	10-11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to explain modern and smart materials and their applications. They will also learn to explain composite materials and technical textiles. Product examples, teacher demonstrations and guided tasks will help them apply the ideas and explain their decisions.

End-of-Week Assessment/Check: How will you know they got it?

Identify a stimulus, response and suitable application. Explain how the material construction improves a property. Review the completed work and final responses to identify misconceptions.

LESSON SLOTS - TUESDAY & WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	15 September 2026	By the end of this lesson, students will be able to explain modern and smart materials and their applications.	**H:** Use a short product example or question to introduce the topic and recall relevant knowledge. **ID:** Demonstrate how a smart material responds to a stimulus and discuss textbook examples. Refer to printed book pages 16-18. **WD:** Complete a second example together. Ask students to explain their decisions and correct misconceptions. **YD:** Match materials to stimuli and uses; justify a material for a product. Record the reasoning using appropriate vocabulary. **C:** Identify a stimulus, response and suitable application.	**Support:** Provide labelled examples, a vocabulary guide and a structured comparison or diagram. Use prompts before independent work. **Extension:** Apply the concept to an unfamiliar product, justify an alternative and explain a trade-off.

2	16 September 2026	By the end of this lesson, students will be able to explain composite materials and technical textiles.	**H:** Use a short product example or question to introduce the topic and recall relevant knowledge. **ID:** Use the book examples to show how combined materials or textile technology improve function. Refer to printed book pages 16-18. **WD:** Complete a second example together. Ask students to explain their decisions and correct misconceptions. **YD:** Compare a composite and a technical textile for stated product needs. Record the reasoning using appropriate vocabulary. **C:** Explain how the material construction improves a property.	**Support:** Provide labelled examples, a vocabulary guide and a structured comparison or diagram. Use prompts before independent work. **Extension:** Apply the concept to an unfamiliar product, justify an alternative and explain a trade-off.
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE Design and Technology Revision Notes, printed pages 16-18
- Teacher computer and classroom display
- Relevant product images, material samples or teacher demonstration models
- Student notebooks, drawing equipment and task sheets



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?